POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2022
Programme: BE | Full Marks: 100
Course: Engineering Mathematics II | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Prove that the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and 4x-3y+1=0=5x-3z+2 are coplanar. Find also the point of their intersection and the plane through them. 7

   b) Find the equation of the sphere through the circle $x^2+y^2+z^2=1$, $2x+4y+5z=6$ and touching the plane $z=0$. 8

2. a) State and prove Euler's theorem on homogeneous function of two independent variables of degree n and 7

   If $u=\cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+\frac{1}{2}\cot u=0.$

   b) Write down the criteria for a function $f(x, y)$ of two variables x & y to have maximum or minimum values at a point. Find the minimum value of $x^2+y^2+z^2$ such that $x+y+z=3a$. 8

3. a) Sketch the region of integration and evaluate the following by interchanging the order of integration. 7

   $$\int_0^2\int_{y^2}^4 y\cos(x^2)\,dx\,dy.$$

   b) Find the volume of the solid whose base is the region in the xy-plane that is bounded by the parabola $y=4-x^2$ and line $y=3x$ while the top of the solid is bounded by the plane $z=x+4$. 8

4. a) Solve $xy'+y = y^2 \log x$ 7

b) Solve. $y''+2y'+y= e^{-x} \cos x$ (method of variation of parameter) 8

5. a) Find the power series solution of a differential equation: $y''-4y =0$. 7

b) Solve the following initial value problem. 8

$$y''+2y'+y = e^{-x};\ y(0) = -1;\ y'(0) = -1$$

6. a) Define Laplace transform of a function and find: 8

i. $L(\sin at \ \cos at)$ Using Laplace transform

ii. $L^{-1}\left\{\dfrac{w}{s^2(s^2+w^2)}\right\}$

b) Using method of Laplace Transformation solve the following initial value problem. 7

$x''-x'-2x=20\sin 2t \quad x(0) = -1,\ x'(0) = 2$

7. Attempt all questions 2.5×4

a) Find radius and center of a sphere $x^2 + y^2 + z^2 + 4x - 6y + 8z = 10$.

b) Find the Laplace transform of $\sin^2 t$.

c) Find the integrating factor of $\dfrac{dy}{dx} + \cot x \ y = \sin x$.

d) If $f(x,y,z) = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$, then show that $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z} = 0$.